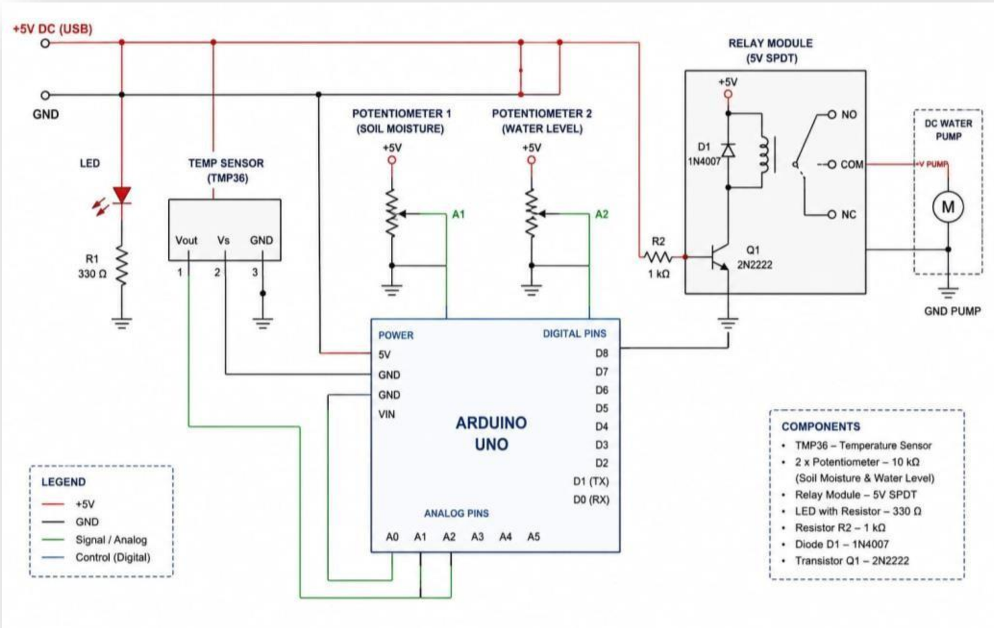
TASK-3

Code:

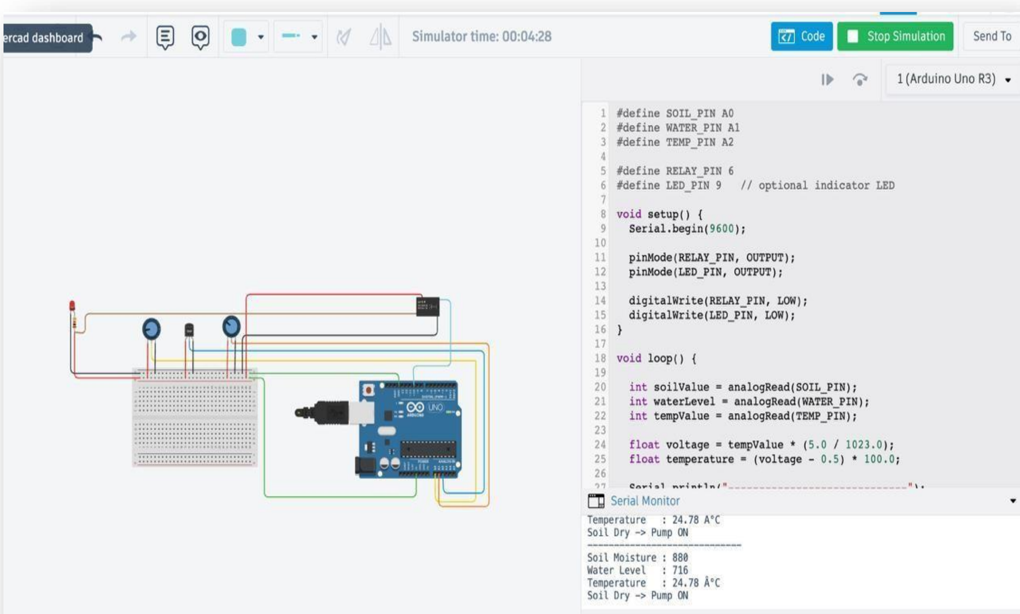
```
#define SOIL_PIN A0
#define WATER_PIN A1
#define TEMP_PIN A2

#define RELAY_PIN 6
#define LED_PIN 9   // optional indicator LED

void setup() {
  Serial.begin(9600);

  pinMode(RELAY_PIN, OUTPUT);
pinMode(LED_PIN, OUTPUT);

  digitalWrite(RELAY_PIN, LOW);
digitalWrite(LED_PIN, LOW);
}

void loop() {

  int soilValue = analogRead(SOIL_PIN);   int
waterLevel = analogRead(WATER_PIN);   int
tempValue = analogRead(TEMP_PIN);   float
voltage = tempValue * (5.0 / 1023.0);   float
temperature = (voltage - 0.5) * 100.0;
```

```
  Serial.println("-----------------------------");

  Serial.print("Soil Moisture : ");
  Serial.println(soilValue);

  Serial.print("Water Level   : ");
  Serial.println(waterLevel);

  Serial.print("Temperature   : ");
  Serial.print(temperature);
  Serial.println(" °C");

  // SPDT Relay control (same logic as normal relay)   if (soilValue > 600) {
digitalWrite(RELAY_PIN, HIGH);  // activates coil → switches COM to NO
digitalWrite(LED_PIN, HIGH);
    Serial.println("Soil Dry -> Pump ON");
  } else {
    digitalWrite(RELAY_PIN, LOW);   // COM switches back to NC
digitalWrite(LED_PIN, LOW);
    Serial.println("Soil Wet -> Pump OFF");
  }

  delay(2000);
}
```

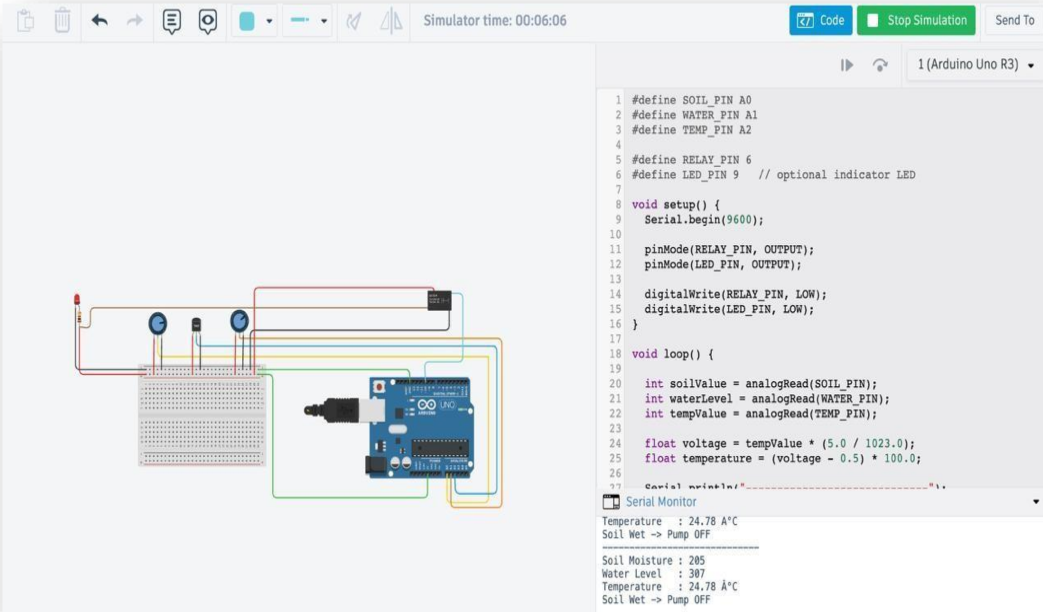
Circuit: **IoT SOIL MONITERING SYSTEM**



Outputs: **POWER ON**



POWER OFF



Simulator time: 00:06:06
Code
Stop Simulation
Send To
1 (Arduino Uno R3)
#define SOIL_PIN A0
#define WATER_PIN A1
#define TEMP_PIN A2
#define RELAY_PIN 6
#define LED_PIN 9 // optional indicator LED
void setup() {
Serial.begin(9600);
pinMode(RELAY_PIN, OUTPUT);
pinMode(LED_PIN, OUTPUT);
digitalWrite(RELAY_PIN, LOW);
digitalWrite(LED_PIN, LOW);
}
void loop() {
int soilValue = analogRead(SOIL_PIN);
int waterLevel = analogRead(WATER_PIN);
int tempValue = analogRead(TEMP_PIN);
float voltage = tempValue * (5.0 / 1023.0);
float temperature = (voltage - 0.5) * 100.0;
Serial Monitor
Temperature : 24.78 Â°C
Soil Wet -> Pump OFF
Soil Moisture : 205
Water Level : 307
Temperature : 24.78 Â°C
Soil Wet -> Pump OFF